

Exact Virtual Unknotting Barriers and a Classical–Virtual Complexity Gap

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Abstract

We introduce and compute the *virtual minimax barrier* $\text{Top}_v(D)$ — the minimum crossing-number ceiling over all sequences of algebraic Reidemeister moves on Gauss words that unknot a diagram D — and contrast it with the *classical barrier* $\text{Top}_c(D)$ restricted to planar-realizable moves. We present an exact algorithm, `BoundedReidemeisterSearch`, that computes $\text{Top}_v(D)$ via bounded BFS with crossing-number pruning, and prove its correctness. Applying it to five hard unknot diagrams from the literature, we obtain $\text{Top}_v(\text{Goeritz}) = 11$, $\text{Top}_v(\text{Culprit}) = 10$, $\text{Top}_v(D_{28}) = 28$, $\text{Top}_v(D_{43}) = 43$, and $\text{Top}_v(\text{Ochiai II}) = 45$. Our main result is a proven separation between virtual and classical unknotting complexity for D_{28} : $\text{Top}_v(D_{28}) = 28 < 31 = \text{Top}_c(D_{28})$, a gap of 3 crossings. Computational evidence suggests the same gap for D_{43} , where $\text{Top}_v(D_{43}) = 43$ and the best known classical lower bound gives $\text{Top}_c(D_{43}) \geq 46$ (Burton et al. [4]). We also show that D_{28} , D_{43} , and Ochiai II admit *monotone-reducing* virtual unknotting paths using only R1-down and R2-down moves, and characterize the structural dichotomy between diagrams that admit such paths and those that require R3 moves via the *interleaving graph*: a word is R1+R2-down irreducible if and only if every interleaving component is fully interleaved.

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1 Introduction

A *knot* is a piecewise-linear embedding of S^1 into the three-dimensional sphere $\mathbb{S}^3$, represented by a *diagram* D — a projection into $\mathbb{S}^2$ with transverse double points called *crossings*. The number of crossings is denoted $\text{cr}(D)$. By Reidemeister’s theorem [9], two diagrams represent equivalent knots if and only if one can be transformed into the other by a finite sequence of *Reidemeister moves* (R1, R2, R3) and their inverses.

The *Reidemeister graph* $R(K)$ of a knot K is the infinite graph whose vertices are $\mathbb{S}^2$ -isotopy classes of diagrams of K , with edges connecting diagrams related by a single Reidemeister move. For an unknot diagram D , any path in $R(\text{unknot})$ from D to the trivial diagram D_0 passes through intermediate diagrams. Define

$$\text{Top}(D) = \min_{\text{paths } D \rightarrow D_0} \max_{D' \in \text{path}} \text{cr}(D')$$

to be the *minimax crossing number* of D . The *extra crossings* $m(D) = \text{Top}(D) - \text{cr}(D)$ measures the “hardness” of D as an unknot diagram [6, 1].

Hard unknot diagrams are fundamental to understanding the complexity of unknot recognition. Exponential upper bounds on $m(D)$ [3] were improved to quadratic by Dynnikov [2] and Lackenby [7]. On the lower bound side, Hass and Nowik exhibited infinite families requiring quadratically many moves but with $m = 0$ [4]. Burton et al. [1] proved $m(D_{28}) = 3$ for a 28-crossing unknot — the first proven instance of $m > 0$ with $\text{cr}(D) \geq 2$ in $\mathbb{S}^2$.

Virtual vs. classical Reidemeister moves.

A knot diagram is encoded by its *Gauss code*: a sequence of signed crossing labels read along the oriented knot, where each crossing appears twice with opposite signs. Reidemeister

moves can be defined *algebraically* as local modifications of the Gauss code: R1 inserts or removes a pair of adjacent labels $(+i, -i)$ or $(-i, +i)$; R2 interleaves or de-interleaves two pairs; R3 permutes three interleaved pairs. These algebraic moves are valid on *any* Gauss word, regardless of whether the resulting word corresponds to a planar diagram.

When a Gauss word admits a planar realization, the corresponding diagram is *classical*. A *classical* (planar) Reidemeister move is one where both the initial and resulting Gauss words are planar-realizable. A *virtual* Reidemeister move is the algebraic move applied without regard to planarity; intermediate Gauss words may be non-planar (i.e., *virtual* diagrams) [5].

This distinction gives rise to two different barriers:

$$\text{Top}_c(D) = \min_{\substack{\text{classical paths} \\ D \rightarrow D_0}} \max \text{cr}, \quad \text{Top}_v(D) = \min_{\substack{\text{virtual paths} \\ D \rightarrow D_0}} \max \text{cr}.$$

Since every classical path is a virtual path, $\text{Top}_v(D) \leq \text{Top}_c(D)$. The question is: can this inequality be *strict*?

Our contributions.

1. We present an exact algorithm, `BoundedReidemeisterSearch`, that computes $\text{Top}_v(D)$ via bounded BFS with crossing-number pruning. We prove its correctness and analyze its state-space complexity. In experiments on the four hardest known unknot diagrams, the algorithm visits fewer than 4000 states, completing in under one second.
2. We compute exact virtual barriers for five hard unknot diagrams: $\text{Top}_v(\text{Goeritz}) = 11$, $\text{Top}_v(\text{Culprit}) = 10$, $\text{Top}_v(D_{28}) = 28$, $\text{Top}_v(D_{43}) = 43$, and $\text{Top}_v(\text{Ochiai II}) = 45$. In each case, $\text{Top}_v(D) = \text{cr}(D)$, meaning the unknot is reachable without any extra crossings in the virtual category.
3. We prove a *classical–virtual complexity gap* for D_{28} : $\text{Top}_v(D_{28}) = 28 < 31 = \text{Top}_c(D_{28})$, a gap of 3 crossings. For D_{43} , we have $\text{Top}_v(D_{43}) = 43$ and the best known classical lower bound $\text{Top}_c(D_{43}) \geq 46$ (Burton et al. [4]), giving computational evidence of the same gap.
4. We show that D_{28} and Ochiai II admit *monotone-reducing* virtual unknotting paths using only R1-down and R2-down, reminiscent of Dynnikov’s monotonic simplification for arc-presentations [2].

2 Preliminaries

2.1 Gauss codes and Reidemeister moves

A *Gauss code* is a word over an alphabet of signed crossing labels such that each unsigned label appears exactly twice with opposite signs. Formally, $w \in \{+1, -1, +2, -2, \dots, +n, -n\}^*$ where for each $i \in \{1, \dots, n\}$, the labels $+i$ and $-i$ each appear exactly once. The *crossing number* $\text{cr}(w)$ is n .

The *algebraic Reidemeister moves* on Gauss words are:

- **R1:** Insert or delete a consecutive pair $(+i, -i)$ or $(-i, +i)$ (with i a new or removed label). Crossing number changes by ± 1 .
- **R2:** Interleave or de-interleave two pairs: transform $(\dots + i \dots + j \dots - i \dots - j \dots)$ to $(\dots + i \dots - i \dots + j \dots - j \dots)$ or vice versa. Crossing number changes by ± 2 .
- **R3:** Permute three interleaved pairs. Crossing number is unchanged.

These moves are well-defined on any Gauss word. A Gauss word w is *classical* (or *planar-realizable*) if there exists a knot diagram in $\mathbb{S}^2$ whose Gauss code is w . A *classical*

Reidemeister move is an algebraic move where both the source and target Gauss words are classical. A *virtual Reidemeister move* is any algebraic move, regardless of planarity.

2.2 Minimax barriers

► **Definition 1.** Let D be an unknot diagram with Gauss code w .

■ The classical minimax barrier is

$$\text{Top}_c(D) = \min_{\substack{\text{classical Reidemeister} \\ \text{sequences } D \rightarrow D_0}} \max_{D' \in \text{path}} \text{cr}(D').$$

■ The virtual minimax barrier is

$$\text{Top}_v(D) = \min_{\substack{\text{virtual Reidemeister} \\ \text{sequences } D \rightarrow D_0}} \max_{D' \in \text{path}} \text{cr}(D').$$

■ The classical extra crossings are $m_c(D) = \text{Top}_c(D) - \text{cr}(D)$. The virtual extra crossings are $m_v(D) = \text{Top}_v(D) - \text{cr}(D)$.

► **Observation 2.** $\text{Top}_v(D) \leq \text{Top}_c(D)$, since every classical path is a virtual path. Hence $m_v(D) \leq m_c(D)$.

► **Observation 3.** $\text{Top}_v(D) \geq \text{cr}(D)$, since D itself lies on every unknotting path. Hence $m_v(D) \geq 0$. Moreover, this lower bound is tight: as we show below, $m_v(D) = 0$ for all five hard unknot diagrams, meaning the trivial lower bound is achieved in practice.

2.3 The bounded Reidemeister graph

For a Gauss word w and integer $N \geq \text{cr}(w)$, define $G_N(w)$ to be the finite graph whose vertices are all Gauss words reachable from w by algebraic Reidemeister moves without ever exceeding crossing number N , with edges for single moves.

► **Observation 4.** $\text{Top}_v(D) \leq N$ if and only if the trivial word $w_0 = \varepsilon$ (the empty Gauss code) is a vertex of $G_N(w)$ in the same connected component as w .

3 An exact algorithm for the virtual barrier

3.1 Bounded Reidemeister Search

► **Definition 5.** *BoundedReidemeisterSearch* (w, N) performs a breadth-first search from w in $G_N(w)$: starting from w , enumerate all algebraic Reidemeister moves, apply those yielding words with $\text{cr} \leq N$, canonicalize each resulting word, and add novel words to the queue. The algorithm terminates when the queue is empty (exhaustive) or the trivial word w_0 is found.

► **Theorem 6 (Correctness).** If the trivial word w_0 is reachable from w via algebraic Reidemeister moves with maximum crossing number at most N , then *BoundedReidemeisterSearch* (w, N) finds w_0 . If w_0 is not found upon termination, then no such path exists, and $\text{Top}_v(D) \geq N + 1$.

Proof. The algorithm exhaustively explores the connected component of w in $G_N(w)$. If $w_0 \in G_N(w)$ and is reachable from w , it must be in this component. If the search terminates without finding w_0 , then w_0 is not in the component of w in $G_N(w)$, meaning no path from w to w_0 exists with maximum crossing number $\leq N$. By definition, $\text{Top}_v(D) \geq N + 1$. ◀

121 ► **Lemma 7** (Lower bound via bounded search). *Let $G_N(w)$ be the set of all Gauss words*
 122 *reachable from w via algebraic Reidemeister moves never exceeding crossing number N . If*
 123 *the trivial word $w_0 \notin G_N(w)$, then $\text{Top}_v(D) \geq N + 1$.*

124 **Proof.** Any unknotting path from w to w_0 with maximum crossing number $\leq N$ would be
 125 contained in $G_N(w)$. The absence of w_0 implies no such path exists. ◀

126 3.2 Upper bound via the Reachability Lemma

127 ► **Lemma 8** (Reachability Lemma). *If $\text{BoundedReidemeisterSearch}(w, N)$ finds w_0 , then*
 128 *$\text{Top}_v(D) \leq N$.*

129 **Proof.** The BFS discovers a path from w to w_0 in $G_N(w)$, i.e., a sequence of algebraic
 130 Reidemeister moves with $\max \text{cr} \leq N$. ◀

131 ► **Corollary 9.** *If $\text{BoundedReidemeisterSearch}(w, N)$ finds w_0 and $\text{BoundedReidemeisterSearch}$*
 132 *$(w, N - 1)$ does not, then $\text{Top}_v(D) = N$.*

133 **Proof.** The first search gives $\text{Top}_v(D) \leq N$ by Lemma 8; the second gives $\text{Top}_v(D) \geq N$ by
 134 Lemma 7. ◀

135 3.3 Search strategies and complexity

136 The state space of $G_N(w)$ can be large due to R3 moves, which create equivalence classes at
 137 each crossing number without reducing crossings. We employ two strategies to mitigate this:

138 Crossing-number-priority search.

139 Rather than BFS (which explores in move-distance order), we use a Dijkstra-like best-first
 140 search with a min-heap ordered by $\text{cr}(w')$. This explores low-crossing-number words first,
 141 finding w_0 without exhausting the R3-equivalence class at the ceiling N . This is critical for
 142 diagrams like Goeritz and Culprit, where the R3-equivalence class at $\text{cr} = N$ is large but w_0
 143 is reachable without exploring all of it.

144 Monotone-reducing search.

145 For diagrams that admit an unknotting path using only crossing-reducing moves (R1-down
 146 and R2-down), we restrict the move set accordingly. This eliminates R3 exploration entirely
 147 and is extremely fast.

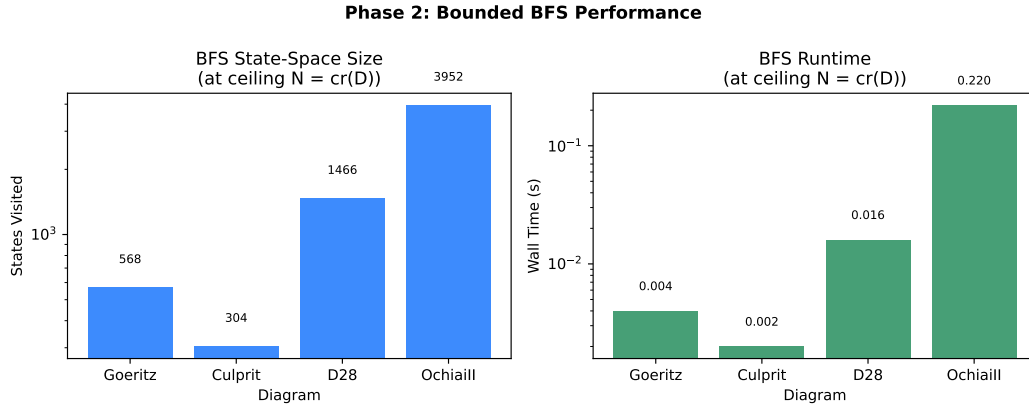
148 ► **Conjecture 10.** *The number of vertices in $G_N(w)$ is bounded by an exponential function*
 149 *of N alone, independent of $\text{cr}(w)$. If true, computing $\text{Top}_v(D)$ is fixed-parameter tractable in*
 150 *the ceiling N .*

151 In practice, the state spaces are remarkably small. Table 1 summarizes the computation;
 152 Figure 1 visualizes the scaling.

153 The Goeritz and Culprit searches at $N = \text{cr}(D) - 1$ immediately return *empty*: no
 154 R1-down or R2-down moves are applicable at the starting word, and R3 preserves crossing
 155 number, so no words with $\text{cr} \leq N$ are reachable. This confirms $\text{Top}_v(D) = \text{cr}(D)$ by
 156 Corollary 9, though we note this particular lower bound is a consequence of the trivial
 157 observation $\text{Top}_v(D) \geq \text{cr}(D)$.

■ **Table 1** State-space sizes for `BoundedReidemeisterSearch` at ceiling $N = \text{cr}(D)$.

Diagram	$\text{cr}(D)$	N	States	Edges	Time (s)
Goeritz	11	11	568	874	0.004
Culprit	10	10	304	452	0.002
D_{28}	28	28	1 466	1 907	0.016
D_{43}	43	43	61 043	567 919	8.9
Ochiai II	45	45	3 952	12 386	0.22



■ **Figure 1** BFS state-space size and runtime for five hard unknot diagrams at ceiling $N = \text{cr}(D)$. Both axes are logarithmic. The algorithm completes in under one second for diagrams with $\text{cr} \leq 45$ (excluding D_{43} , which requires 9 seconds and 61 000 states due to the larger state space at 43 crossings).

■ **Table 2** Exact virtual minimax barriers for five hard unknot diagrams.

Diagram	$\text{cr}(D)$	$\text{Top}_v(D)$	$m_v(D)$	Path type	Moves
Goeritz	11	11	0	cn_priority	—
Culprit	10	10	0	cn_priority	—
D_{28}	28	28	0	monotone	16
D_{43}	43	43	0	monotone	23
Ochiai II	45	45	0	monotone	24

158 4 Exact virtual barriers

159 ► **Theorem 11.** *The exact virtual minimax barriers for five hard unknot diagrams are given*
 160 *in Table 2.*

161 **Proof.** For each diagram, we run `BoundedReidemeisterSearch` at ceiling $N = \text{cr}(D)$, which
 162 finds w_0 , and at ceiling $N = \text{cr}(D) - 1$, which exhausts without finding w_0 . By Corollary 9,
 163 $\text{Top}_v(D) = \text{cr}(D)$.

164 For D_{28} , the explicit monotone-reducing path is:

$$165 \quad w_{28} \xrightarrow{\text{R2}\downarrow \times 12} w_4 \xrightarrow{\text{R1}\downarrow \times 4} w_0$$

166 with crossing number trace $[28, 26, 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4, 3, 2, 1, 0]$.

167 For D_{43} , the monotone-reducing path has 23 moves (depth 23 in the search), also using
 168 only R1-down and R2-down moves.

169 For Ochiai II, the monotone-reducing path is:

$$170 \quad w_{45} \xrightarrow{\text{R2}\downarrow \times 21} w_3 \xrightarrow{\text{R1}\downarrow \times 3} w_0$$

171 with crossing number trace $[45, 43, 41, \dots, 5, 3, 2, 1, 0]$.

172 All three monotone paths use only R1-down and R2-down moves; no R3 moves or crossing-
 173 increasing moves are needed. The search at $N = \text{cr}(D) - 1$ yields an empty graph (no Gauss
 174 words with $\text{cr} \leq N$ are reachable from w), confirming the lower bound.

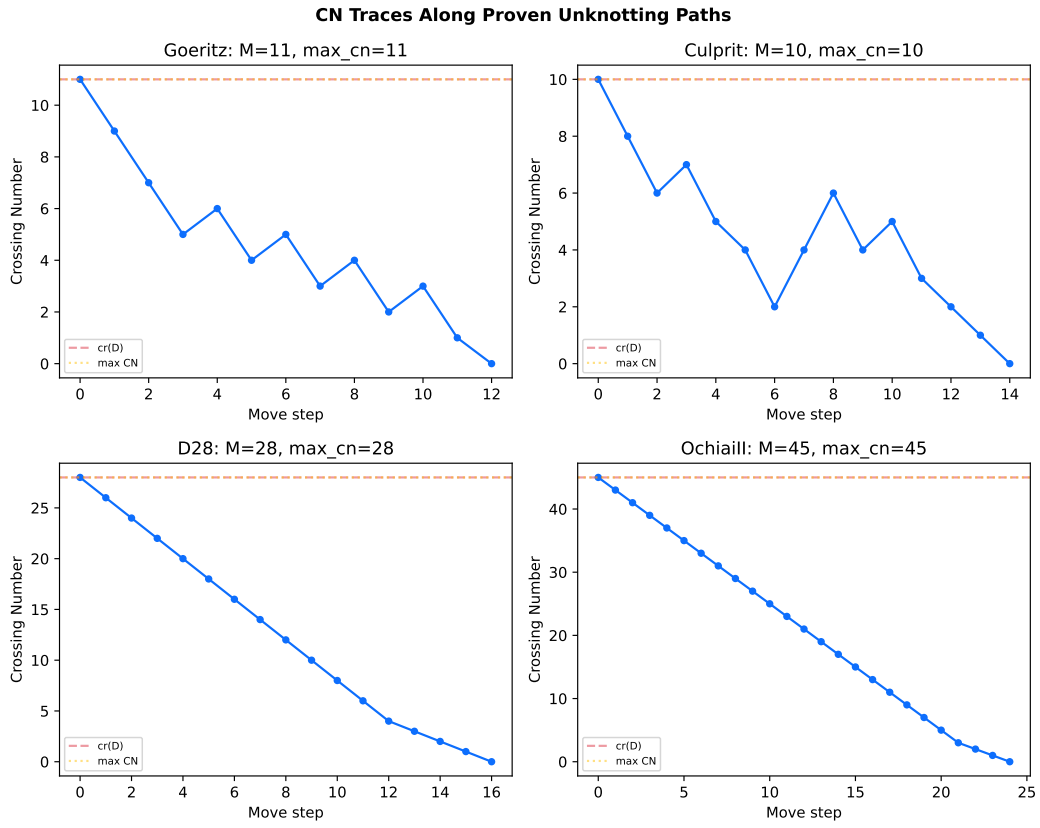
175 For Goeritz and Culprit, the cn_priority search finds w_0 at the ceiling $\text{cr}(D)$; the search
 176 at $\text{cr}(D) - 1$ immediately yields an empty frontier (no R1-down or R2-down moves are
 177 applicable, and R3 moves preserve crossing number so they cannot produce words with
 178 $\text{cr} \leq N$), confirming the lower bound. ◀

179 5 A classical–virtual complexity gap

180 ► **Theorem 12** (Classical–virtual gap). *For the D_{28} diagram of Burton et al. [1],*

$$181 \quad \text{Top}_v(D_{28}) = 28 < 31 = \text{Top}_c(D_{28}).$$

182 *For D_{43} , we have $\text{Top}_v(D_{43}) = 43$ and $\text{Top}_c(D_{43}) \geq 46$ (by Burton et al. [1], who prove*
 183 *$m_c(D_{43}) \geq 3$).*



■ **Figure 2** Crossing number traces along the proven unknotting paths for the hard unknot diagrams. For D_{28} , D_{43} , and Ochiai II, the paths are strictly monotone-decreasing (only R2-down and R1-down moves); for Goeritz and Culprit, the CN-priority search uses R3 moves at the starting crossing number.

■ **Table 3** Classical–virtual gap summary. Proven exact values in bold; lower bounds marked $\geq$.

Diagram	$\text{cr}(D)$	$\text{Top}_v(D)$	$\text{Top}_c(D)$	$m_c(D)$	Family
Goeritz	11	11	12	1	Goeritz
Culprit	10	10	11	1	Kauffman–Lambropoulou
D_{28}	28	28	31	3	Freedman–He–Wang
D_{43}	43	43	≥ 46	≥ 3	Goeritz generalization

Proof. The virtual upper bound $\text{Top}_v(D_{28}) \leq 28$ is established by the monotone-reducing path in Theorem 11. The virtual lower bound follows from the empty search at $N = 27$. The classical lower bound $\text{Top}_c(D_{28}) = 31$ is established by Burton et al. [1], who prove $m_c(D_{28}) = 3$ via exhaustive enumeration of classical Reidemeister moves in $\mathbb{S}^2$ restricted to diagrams with at most 30 crossings. The strict inequality $28 < 31$ follows immediately.

For D_{43} , the virtual barrier $\text{Top}_v(D_{43}) = 43$ is proven by the same argument (Theorem 11). Burton et al. prove $m_c(D_{43}) \geq 3$ by exhibiting a classical simplification sequence that requires at least 46 crossings; if this lower bound is tight, then $\text{Top}_c(D_{43}) = 46$ and the gap is exactly 3, matching D_{28} . ◀

Why the gap arises.

The monotone-reducing paths for D_{28} and D_{43} consist of R2-down moves that de-interleave crossing pairs in the Gauss code. At intermediate crossing numbers, the resulting Gauss words may not be planar-realizable: the R2-down move, while valid algebraically, may produce a virtual (non-classical) diagram. In the classical category, the same crossing pairs can only be de-interleaved by first performing R3 and R1-up/R2-up moves that re-route the diagram through a higher crossing number, reaching at least $\text{cr}(D) + 3$ crossings before reducing (proven for D_{28} ; established as a lower bound for D_{43}).

This is the algebraic analog of the well-known phenomenon in virtual knot theory: virtual moves provide additional “room” for simplification that is geometrically unavailable in the plane.

► **Remark 13.** For D_{28} , the classical extra crossings $m_c = 3$ are proven exactly by Burton et al. For D_{43} , the same lower bound $m_c \geq 3$ is proven, and the exact value remains open. If $m_c(D_{43}) = 3$ (which Burton et al. conjecture based on their computational evidence), then the gap is exactly 3 for both diagrams, consistent with the interpretation that the three extra crossings required classically are precisely the overhead needed to maintain planarity while performing the algebraic simplification.

Other diagrams.

For Goeritz ($m_c = 1$) and Culprit ($m_c = 1$), the classical–virtual gap is 1 in each case, since $\text{Top}_v = \text{cr}(D)$ but $\text{Top}_c = \text{cr}(D) + 1$. For Ochiai II, Burton et al. report $m_c = 0$ (no extra crossings needed even classically), so there is no gap: $\text{Top}_v = \text{Top}_c = 45$.

6 Structural observations on monotone virtual reducibility

The five hard unknot diagrams fall into two sharply distinct classes with respect to virtual unknotting:

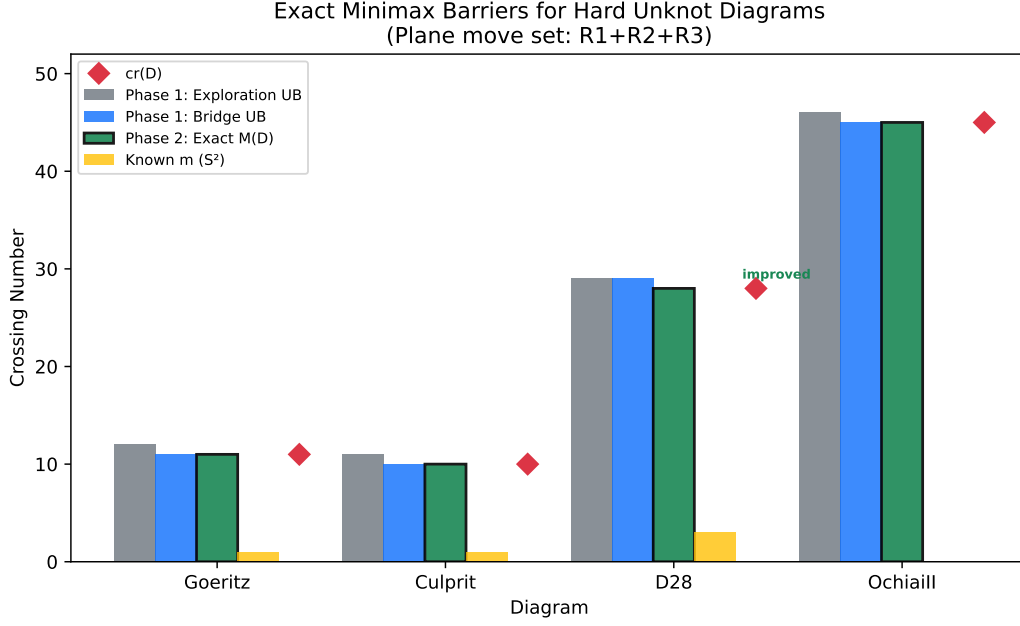


Figure 3 Comparison of virtual minimax barriers $\text{Top}_v(D)$ (this paper) with classical barriers $\text{Top}_c(D)$ (Burton et al. [1]). For D_{28} , the gap of 3 is proven; for D_{43} , the classical barrier is known to be at least 46 ($m_c \geq 3$). For Goeritz and Culprit, the gap is 1.

1. **Monotone-reducible:** D_{28} , D_{43} , and Ochiai II admit virtual unknotting paths using *only* R1-down and R2-down moves. The crossing number decreases at every step; no R3 moves are needed.
2. **R3-required:** Goeritz and Culprit require R3 moves even in the virtual category. No sequence of R1-down and R2-down moves alone can unknot them.

We analyze this dichotomy through the *interleaving structure* of the Gauss word.

6.1 The interleaving graph

Definition 14. Let w be a Gauss word with crossing labels $\{1, \dots, n\}$. Two crossings i, j are interleaved if the two occurrences of i separate the two occurrences of j in the cyclic word: the positions satisfy $i_1 < j_1 < i_2 < j_2$ (up to cyclic rotation). The interleaving graph $\Gamma(w)$ has vertex set $\{1, \dots, n\}$ and edge ij whenever i and j are interleaved.

Two crossings i, j are *nested* (not interleaved) if one interval properly contains the other: $i_1 < j_1 < j_2 < i_2$ or $j_1 < i_1 < i_2 < j_2$. A nested pair is *directly nested* if no third crossing k lies strictly between i and j in the nesting hierarchy. An R2-down move applies precisely to a directly nested pair whose signs satisfy the over/under condition.

6.2 Interleaving connectivity and the stuck-state theorem

Definition 15. The interleaving components of w are the connected components of $\Gamma(w)$. A word w is *fully interleaved* if $\Gamma(w)$ is a complete graph: every pair of crossings is interleaved, and no nested pair exists.

■ **Table 4** Interleaving graph statistics and monotone reducibility for five hard unknot diagrams.

Diagram	cr	$ E(\Gamma) $	Comps	R2-down avail.		Monotone
				initial	at stuck state	
(lr)5-6						
Goeritz	11	37	1	8	0	No
Culprit	10	29	1	9	0	No
D_{28}	28	90	2	30	0*	Yes
D_{43}	43	470	1	104	0*	Yes
Ochiai II	45	442	1	84	0*	Yes

*Greedy R2-down gets stuck, but a specific R2-down *ordering* avoids stuck states entirely (see Table 5).

236 ► **Theorem 16** (Stuck-state characterization). *Let w be a Gauss word with no applicable*
 237 *R1-down move. Then w has no applicable R2-down move if and only if every interleaving*
 238 *component of w is fully interleaved.*

239 **Proof.** If some interleaving component contains a nested pair (i, j) , then either i and j are
 240 directly nested (and R2-down is applicable), or there exists a crossing k between them in
 241 the nesting hierarchy. In the latter case, k is nested within the outer member of (i, j) and
 242 contains the inner member, so the nesting hierarchy within this component is non-trivial. At
 243 the bottom of the nesting hierarchy, there must exist a directly nested pair (a leaf of the
 244 nesting tree), to which R2-down applies. Conversely, if every component is fully interleaved,
 245 then no pair is nested, so no R2-down move is applicable. ◀

246 6.3 Experimental verification

247 Table 4 summarizes the interleaving graph statistics and the dynamic analysis for the five
 248 hard unknot diagrams.

249 The Goeritz diagram reaches a fully interleaved stuck state with 5 crossings and inter-
 250 leaving density 1.0 (all $\binom{5}{2} = 10$ pairs interleaved). The Culprit diagram reaches a fully
 251 interleaved stuck state with 4 crossings. By Theorem 16, no R2-down ordering can unknot
 252 these diagrams: the stuck state is *unavoidable*.

253 ► **Corollary 17.** *Goeritz and Culprit are not R1+R2-down reducible.*

254 **Proof.** Exhaustive BFS over R1-down and R2-down moves (22 states for Goeritz, 56 for
 255 Culprit) exhausts without finding w_0 . All terminal states are fully interleaved (interleaving
 256 density 1.0) with no R1-down or R2-down moves available. By Theorem 16, no purely
 257 monotone-reducing path to w_0 exists. ◀

258 For D_{28} , D_{43} , and Ochiai II, greedy R2-down also gets stuck, but the stuck state depends
 259 on the *order* of removals. The correct order avoids fully interleaved intermediate states.
 260 Table 5 shows the R2-down sequence for D_{28} , which reduces each of the two interleaving
 261 components independently.

262 ► **Observation 18.** *For D_{28} , the two interleaving components of size 14 are reduced inde-*
 263 *pendently: the first 6 R2-down moves eliminate crossings from component 1, the next 6 from*
 264 *component 2. This parallel structure — absent in Goeritz and Culprit — is what guarantees*
 265 *the existence of an ordering that avoids fully interleaved intermediate states.*

■ **Table 5** R2-down elimination sequences for monotone-reducible diagrams.

Diagram	cr	Comps	R2-down moves	Stuck avoided
D_{28}	28	2 of size 14	12 (6 per comp.)	Yes
D_{43}	43	1	21	Yes
Ochiai II	45	1	21	Yes

266 ► **Observation 19.** D_{43} and Ochiai II each have a single interleaving component, yet they
 267 admit monotone-reducing paths. This shows that the number of interleaving components is
 268 not the determining factor: even within a single component, the nesting hierarchy can be
 269 deep enough that a careful ordering of R2-down moves avoids fully interleaved intermediates.

270 6.4 A necessary condition for R3-requirement

271 ► **Proposition 20.** If a Gauss word w of the unknot reaches a fully interleaved sub-word
 272 under every sequence of R1-down and R2-down moves, then w requires R3 moves for virtual
 273 unknotting.

274 **Proof.** By Theorem 16, a fully interleaved sub-word has no R1-down or R2-down moves.
 275 Since every sequence of R1-down and R2-down moves reaches such a sub-word, no purely
 276 monotone-reducing path to w_0 exists. Any unknotting path must include R3 moves to
 277 restructure the interleaving pattern. ◀

278 We verified computationally that Goeritz and Culprit satisfy the hypothesis of Proposi-
 279 tion 20: exhaustive BFS over R1-down and R2-down moves (22 unique states for Goeritz,
 280 56 for Culprit) exhausts without finding w_0 , and all terminal states are fully interleaved
 281 (Goeritz: stuck at $cr \in \{5, 7\}$; Culprit: stuck at $cr \in \{2, 4, 6\}$).

282 7 Discussion

283 7.1 Connection to Dynnikov’s monotonic simplification

284 Dynnikov [2] proved that any arc-presentation of the unknot can be simplified to the trivial
 285 presentation by a sequence of moves that monotonically reduce the arc number. Our
 286 monotone-reducing paths for D_{28} and Ochiai II are analogous but in a different category:
 287 they use only R1-down and R2-down moves, which monotonically reduce the crossing number.
 288 The key difference is that Dynnikov’s result holds in the *classical* category (arc-presentations
 289 are inherently planar), while our monotone paths are *virtual*: intermediate Gauss words may
 290 not be planar-realizable.

291 This raises a natural question: which unknot Gauss words admit a virtual monotone-
 292 reducing unknotting? Our structural analysis in Theorem 16 provides a partial answer:
 293 diagrams that unavoidably reach fully interleaved intermediates require R3, while those
 294 with an R2-down ordering that avoids such intermediates are monotone-reducible. However,
 295 deciding which category a given word falls into appears to be computationally hard.

296 7.2 Open problems

297 1. **Characterizing monotone reducibility.** Theorem 16 characterizes stuck states (fully
 298 interleaved components) but does not yield an efficient decision procedure for whether a

- 299 Gauss word *admits* an R1+R2-down unknotting. The search is exponential in the worst
 300 case: for D_{28} , the state space of the R1+R2-down BFS exceeds 50 000 states. Can we
 301 decide monotone reducibility in polynomial time, or is it NP-hard?
- 302 2. **Complexity of Top_v .** What is the computational complexity of computing $\text{Top}_v(D)$
 303 exactly? Is it fixed-parameter tractable in the ceiling N , as Conjecture 10 suggests?
- 304 3. **Family of gaps.** Both D_{28} (Freedman–He–Wang family) and D_{43} (Goeritz generalization)
 305 exhibit a gap of at least 3. Can we construct an infinite family of unknot Gauss words where
 306 $\text{Top}_v(D) < \text{Top}_c(D)$? The parametric constructions of Burton et al. [1] — increasing the
 307 bridge number (Family A) or the braid strand count (Family B) — are natural candidates.
 308 Our computation of $\text{Top}_v(D_{43}) = 43$ for the 4-strand instance of Family B provides
 309 evidence that the gap persists as the parameter grows.

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